

Quasimodular forms for the orthogonal group and Gromov-Witten theory

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based mostly on arXiv:2505.09535 with Brandon Williams
and a bit on upcoming work with Pandharipande

Plan of the talk

- Motivation
- Definition of quasimodular forms for the orthogonal group
- Theta lift and quasimodular forms
- Back to geometry

Enriques surfaces

An Enriques surface is a smooth projective surface Y with non-trivial canonical bundle such that

$$\omega_Y^{\otimes 2} \cong \mathcal{O}_Y, \quad H^1(Y, \mathcal{O}_Y) = 0.$$

Hodge diamond:

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 10 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

Intersection lattice:

$$H^2(Y, \mathbb{Z})/\text{torsion} = U \oplus E_8(-1).$$

The double cover associated to ω_Y is a K3 surface X , so $Y = X/\mathbb{Z}_2$. Moduli of Enriques surfaces is isomorphic to moduli space of K3 surfaces with lattice polarization $U(2) \oplus E_8(-2)$.

Gromov-Witten theory

For $\beta \in H^2(Y, \mathbb{Z})$, consider the GW invariants of the (local) Enriques surface:

$$N_{g,\beta} = \int_{[\overline{M}_g(Y,\beta)]^{\text{vir}}} (-1)^{g-1} \lambda_{g-1}$$

Form the generating series

$$F_g(Q) = \sum_{\beta \geq 0} N_{g,\beta} Q^\beta.$$

The starting point of this investigation was the following theorem:

Theorem (O. 2023)

For any $g \geq 1$ we have

$$F_g = 2(-1)^{g-1} \sum_{\beta > 0} Q^\beta \sum_{\text{odd } k|\beta} k^{2g-3} \omega_g \left(\frac{\beta^2}{2k^2} \right)$$

where the coefficients $\omega_g(n)$ are defined by

$$\sum_{g \geq 0} \sum_{n \geq 0} \omega_g(n) z^{2g-2} q^n = \prod_{\substack{m \geq 1 \\ m \text{ odd}}} \frac{1}{(1 - e^z q^m)^2 (1 - e^{-z} q^m)^2 (1 - q^m)^4} \prod_{m \geq 1} \frac{1}{(1 - q^m)^8}.$$

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Question

- What is the modular interpretation of the series F_g ?
- What can we expect for K3 surfaces?

Theorem (Dijkgraaf, Kaneko-Zagier, Okounkov-Pandharipande, ...)

Gromov-Witten invariants of an elliptic curve are quasimodular forms for $SL_2(\mathbb{Z})$.

Physics/Mirror symmetry (Grimm-Klemm-Marino-Weiss): F_g should be the asymptotic expansions of (mildly) non-holomorphic modular objects on the moduli space of Enriques surfaces.

The moduli space of elliptic curves parametrize weight 1 Hodge structures.
The moduli space of K3/Enriques surfaces parametrize weight 2 Hodge structures.

→ Need quasimodular forms for the orthogonal group!

Quasimodular forms: Classical case

Let $\mathbb{H} = \{\tau = x + iy \in \mathbb{C} \mid y > 0\}$. A function $f : \mathbb{H} \rightarrow \mathbb{C}$ is *almost-holomorphic* if f is a polynomial in $1/y$ with coefficients holomorphic functions.

Definition

A quasi-modular form of weight k for $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ which is the constant term of a almost-holomorphic modular form F of weight k for Γ .

- $\mathrm{ct} : \mathrm{AHMod}_k(\Gamma) \rightarrow \mathrm{QMod}_k(\Gamma)$ "constant-term map" is an isomorphism
- $\mathrm{QMod}(\Gamma) = \mathrm{Mod}(\Gamma)[G_2]$ where

$$G_2(\tau) = -\frac{1}{24} + \sum_{n \geq 1} \sum_{d \mid n} dq^n$$

- $D_\tau = q \frac{d}{dq}$ acts on $\mathrm{QMod}(\Gamma)$ with weight $+2$.
- There is the natural operator

$$\frac{d}{dG_2} : \mathrm{QMod}(\Gamma) \rightarrow \mathrm{QMod}(\Gamma)$$

such that $D_\tau, \frac{d}{dG_2}$, wt is an $\mathfrak{sl}_2$ -triple.

We can consider the Maaßraising and lowering operators on $C^\infty(\mathbb{H})$:

$$R_k f = 2i \frac{\partial}{\partial \tau} f + \frac{k}{y} f, \quad L_k f = -2iy^2 \frac{\partial}{\partial \bar{\tau}} f.$$

Then f is almost-holomorphic if and only if $\exists m \geq 1$ such that $L^m f = 0$.

How to find L_k, R_k ?

This is a general problem:

Let $\mathcal{E}$ be a holomorphic vector bundle on a complex manifold X .

Let $U \subset X$ be some open with coordinates $x_1, \dots, x_n$ such that $\mathcal{E}|_U \cong U \times \mathbb{C}^r$.

Let s be a section of $\mathcal{E}$. Then

$$\partial(s) = \sum_i \left(\frac{\partial s}{\partial x_i} \right) dx_i$$

does not define a section of $\mathcal{E} \otimes \Omega_M$ since it depends choice of trivialization.

Instead we want to define a connection

$$R : \mathcal{A}(M, \mathcal{E}) \rightarrow \mathcal{A}(M, \mathcal{E} \otimes \Omega_M)$$

satisfying the Leibniz rule $R(fs) = \partial(f)s + fR(s)$.

Let H be a Hermitian metric on $\mathcal{E}$, i.e. a Hermitian inner product on $\mathcal{E}|_x$ for all $x \in M$. We obtain an isomorphism

$$\psi : \mathcal{E} \cong \overline{\mathcal{E}^*}, v \mapsto \overline{H(v, -)}.$$

Since $\overline{\mathcal{E}^*}$ is anti-holomorphic, ∂ is defined as a global derivative.

Define

$$R := (\psi^{-1} \otimes \text{id}_{\Omega_M}) \circ \partial \circ \psi : \mathcal{A}(M, \mathcal{E}) \rightarrow \mathcal{A}(M, \mathcal{E} \otimes \Omega_M).$$

Construction on $\mathbb{H}$

On $\mathbb{H}$ we have the ($\mathrm{SL}_2(\mathbb{R})$ -equivariant) Hodge line bundle $\mathcal{L}$ (so that a modular form of weight k corresponds to a section of $\mathcal{L}^k$).

It has a canonical $\mathrm{SL}_2(\mathbb{R})$ -invariant Hermitian (Kähler) form

$$\omega = -2i\partial\bar{\partial}\log(y)$$

which is a section of $\mathcal{L}^{\otimes 2} \otimes \overline{\mathcal{L}}^2$ and gives $\mathrm{SL}_2(\mathbb{R})$ -equivariant isomorphism

$$\mathcal{L}^{\otimes 2} \xrightarrow{\cong} \overline{\mathcal{L}^*}^{\otimes 2}.$$

Then

$$R_k = \left[\mathcal{L}^{2k} \xrightarrow{\omega^k} \overline{\mathcal{L}^{2k*}} \xrightarrow{\partial} \overline{\mathcal{L}^{2k*}} \otimes \mathcal{L}^2 \xrightarrow{\omega^{-k} \otimes \mathrm{id}_{\mathcal{L}^2}} \mathcal{L}^{2k} \otimes \mathcal{L}^2 \cong \mathcal{L}^{2k+2} \right].$$

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In essence this is the general construction behind Shimura's "nearly-holomorphic functions" defined for any Kähler manifold:

- Define raising and lowering operators using the Kähler metric.
- Define almost-holomorphic (holomorphic) as functions killed by $L^{\gg 0}$.

Let M be an even integral lattice of signature $(2, n)$ and let

$$\mathcal{D} = \mathcal{D}_M = \{[Z] \in \mathbb{P}(M_{\mathbb{C}}) \mid Z \cdot Z = 0, Z \cdot \bar{Z} > 0\}^+$$

Let $O^+(M) \leq O(M)$ be the subgroup preserving the $+$ component.

Orthogonal modular forms

Let M be an even integral lattice of signature $(2, n)$ and let

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Two Hodge bundles:

- $\mathcal{L} = \mathcal{O}_{\mathbb{P}(M_{\mathbb{C}})}(-1)|_{\mathcal{D}}$ (a subbundle of $M_{\mathbb{C}} \otimes \mathcal{O}_{\mathcal{D}}$)
- $\mathcal{E} = \mathcal{L}^{\perp} / \mathcal{L}$ ($\mathcal{E}$ is an orthogonal rank n bundle, i.e. $\mathcal{E} \cong \mathcal{E}^{\vee}$).

The cotangent bundle is

$$\Omega_{\mathcal{D}} = \mathcal{E}^{\vee} \otimes \mathcal{L} \cong \mathcal{E} \otimes \mathcal{L}.$$

Let Γ be a finite index subgroup of $O^+(M)$.

Definition

A modular form of weight k and rank s for Γ is a holomorphic section of $\mathcal{L}^{\otimes k} \otimes \mathcal{E}^{\otimes s}$ which is invariant under Γ .

Raising operators

There is a $O^+(M)$ -invariant Hermitian inner product on $\mathcal{T}_{\mathcal{D}}$ given by

$$\omega = \partial\bar{\partial} \log(\operatorname{Im}(z)^2)$$

Moreover, there is a Hermitian metric on $\mathcal{L}$ given by

$$|Z|^2 \in \mathcal{A}(\mathcal{D}, \mathcal{L}^* \otimes \overline{\mathcal{L}^*}).$$

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This defines a morphism

$$\begin{aligned}\psi &:= |Z|^2 \omega : \mathcal{E} \rightarrow \overline{\mathcal{E}} \\ |Z|^{2k} \psi^{\otimes s} &: \mathcal{L}^{\otimes k} \otimes \mathcal{E}^{\otimes s} \rightarrow (\overline{\mathcal{L}^*})^k \otimes (\overline{\mathcal{E}^*})^{\otimes s}\end{aligned}$$

Definition

On any open $U \subset \mathcal{D}$ the orthogonal lowering and raising operators are

$$L_k := |Z|^2 (\operatorname{id}_{\mathcal{E}^{\otimes s}} \otimes \varphi) \circ \bar{\partial} : \mathcal{A}(U, \mathcal{L}^k \otimes \mathcal{E}^s) \rightarrow \mathcal{A}(U, \mathcal{L}^{k-1} \otimes \mathcal{E}^{s+1});$$

$$R_k := |Z|^{-2k} (\varphi^{\otimes s} \otimes \operatorname{id}_{\mathcal{E}}) \circ \partial \circ |Z|^{2k} \psi^{\otimes s} : \mathcal{A}(U, \mathcal{L}^k \otimes \mathcal{E}^s) \rightarrow \mathcal{A}(U, \mathcal{L}^{k+1} \otimes \mathcal{E}^{s+1})$$

Definition

A section $F \in \mathcal{A}(U, \mathcal{E}^{\otimes s} \otimes \mathcal{L}^k)$ is called *almost-holomorphic of depth d* if $L^d(F) \neq 0$ and $L^{d+1}(F) = 0$. (This defines almost-holomorphic modular forms!)

Tube domain and quasimodular forms

Assume that $M = L \stackrel{\perp}{\oplus} U$. Consider the tube domain

$$\mathcal{C} = \{z \in L_{\mathbb{C}} | \operatorname{Im}(z)^2 > 0\}^+$$

There is a biholomorphism

$$\varphi : \mathcal{C} \xrightarrow{\cong} \mathcal{D}, \quad \varphi(z) = \left(-\frac{1}{2}z^2, z, 1\right). \quad (1)$$

One finds that a section s is almost-holomorphic iff its coefficients are polynomials in the non-holomorphic variables

$$\nu_i = \partial_{z_i} \log(\operatorname{Im}(z)^2)$$

We also fix a trivialization $\mathcal{E}|_{\mathcal{C}} = \mathcal{C} \times \mathbb{C}^n$ naturally.

Definition

A function $f = (f_{i_1, \dots, i_s}) : \mathcal{C} \rightarrow (\mathbb{C}^{\otimes n})^s$ is a quasimodular form of weight k and rank s for Γ if there exists a almost-holomorphic modular form $F = F_{i_1, \dots, i_s}$ such that

$$F_{i_1, \dots, i_s} |_{\nu_1 = \dots = \nu_n = 0} = f.$$

Theorem (O-W)

The constant-term morphism

$$\text{ct} : \text{AHMod}_{k,s}(\Gamma) \rightarrow \text{QMod}_{k,s}(\Gamma), \quad F \mapsto F|_{\nu=0},$$

is an isomorphism.

Theorem (O-W)

The lowering and raising operators act on the spaces of almost-holomorphic modular forms as derivations by

$$L_k : \text{AHMod}_{k,s}(\Gamma) \rightarrow \text{AHMod}_{k-1,s+1}(\Gamma), \quad R_k : \text{AHMod}_{k,s}(\Gamma) \rightarrow \text{AHMod}_{k+1,s+1}(\Gamma)$$

This gives maps on quasimodular forms

$$L_k : \text{QMod}_{k,s}(\Gamma) \rightarrow \text{QMod}_{k-1,s+1}(\Gamma), \quad R_k : \text{QMod}_{k,s}(\Gamma) \rightarrow \text{QMod}_{k+1,s+1}(\Gamma)$$

These operators satisfy a version of " $[L_k, R_k] = \text{wt}$ ". On quasimodular forms R_k is just the derivative ∂ .

Analogues of $G_2(q)$

Recall that

$$G_2(q) = \frac{D_\tau \eta(\tau)}{\eta(\tau)}, \quad G_2^*(q) = \frac{R(\eta)}{\eta}.$$

To introduce an analogue for orthogonal modular form we will have to consider quotient as well, so one needs to allow poles along the associated divisor.

Concretely, given any modular form f of weight $\ell > 0$ on $\mathcal{D}$ for Γ , an analogue of G_2^* is defined by

$$G := \frac{1}{\ell} \frac{\partial(f)}{f}, \quad G^* := \frac{1}{\ell} \frac{R(f)}{f}$$

Consider the vanishing locus of f :

$$\mathcal{H} = \{Z \in \mathcal{D} \mid f(Z) = 0\}.$$

Definition

A *weak modular form* on $(\mathcal{D}, \mathcal{H})$ of weight k and rank s for Γ is a meromorphic Γ -invariant section of $\mathcal{L}^{\otimes k} \otimes \mathcal{E}^{\otimes s}$ which has a pole of order $\leq k$ along $\mathcal{H}$ and no other poles. We let $\text{Mod}_{k,s}^w(\Gamma)$ be the vector space of such forms.

Definition

A weak almost-holomorphic modular form on $(\mathcal{D}, \mathcal{H})$ of weight k and rank s for Γ is a section $F \in \mathcal{A}(\mathcal{D} \setminus \mathcal{H}, \mathcal{L}^k \otimes \mathcal{E}^{\otimes s})$ such that:

- (i) F is almost-holomorphic;
- (ii) F is Γ -invariant: $\gamma^*(F) = F$ for all $\gamma \in \Gamma$;
- (iii) $L^d(F)$ has a pole of order $\leq k - d$ along $\mathcal{H}$ for all $d \geq 0$.

The notion of weak (almost-holomorphic/quasi) modular forms seems to be the right setting for all kinds of structure results. For example, we have an analogue of

$$\mathrm{QMod}(\Gamma) = \mathrm{Mod}(\Gamma)[G_2].$$

Theorem (OW)

We have a natural isomorphism

$$\mathrm{AHMod}_{k,s}^{\mathbf{w}}(\Gamma) \cong \bigoplus_{d=0}^k \mathrm{Mod}_{k-d,s+d}^{\mathbf{w}}(\Gamma)^{\mathrm{Sym}_d}$$

where $\mathrm{Mod}_{k-d,s+d}^{\mathbf{w}}(\Gamma)^{\mathrm{Sym}_d} \subset \mathrm{Mod}_{k-d,s+d}^{\mathbf{w}}(\Gamma)$ is the subspace of rank $s + d$ forms invariant under the S_d action permuting the last d factors.

Let F be an almost-holomorphic vector-valued modular form for the finite Weil representation of M of weight $\kappa = k + 1 - \frac{n}{2}$ (with no poles in the interior but with possibly a pole at ∞ with at worst exponential growth.)

For $k \geq 0$ the regularized theta lift is

$$\text{Lift}(F)(Z) = \frac{1}{2} \left(\frac{i}{2} \langle Z, \overline{Z} \rangle \right)^{-k} \int^{\text{reg}} [F(\tau), \Theta_k(\tau, Z)] \frac{dx dy}{y^2}.$$

and transforms like a modular form of weight k under the discriminant kernel

$$\tilde{O}^+(M) = \text{Ker}(O^+(M) \rightarrow O(M^\vee/M)).$$

Here

$$\Theta_k(\tau, Z) = y^{n/2} \sum_{\lambda \in M^\vee} \langle \lambda, Z \rangle^k e^{\pi i \tau \langle \lambda_Z, \lambda_Z \rangle + \pi i \overline{\tau} \langle \lambda_Z^\perp, \lambda_Z^\perp \rangle} e_{\lambda+M}.$$

is a certain theta function.

Interaction with Lowering operators

A key result that we prove is the following:

Theorem

Suppose F is an almost-holomorphic modular form of weight $\kappa = k + 1 - n/2$ with $k \geq 2$. The image of $\text{Lift}(F)$ under the orthogonal lowering operator is

$$L_k[\text{Lift}(F)] = -\frac{1}{2\pi} \cdot R_{k-2}[\text{Lift}(L^{\text{Maa}\beta} F)].$$

This gives an if-and-only-if criterion for when a lift of an almost-holomorphic modular form is again almost-holomorphic.

Theorem

Let F be an almost-holomorphic modular form of weight $\kappa = k + 1 - \frac{n}{2}$ and depth $d > 0$.

- If $k \geq 2d$, then $\text{Lift}(F)$ is a weak almost-holomorphic modular form of weight k and depth $2d$.*
- If in addition M splits as $M = U \oplus L$, then $\text{Lift}(F)$ is almost-holomorphic if and only if $k \geq 2d$.*

Examples

Consider $M = U \oplus U$ and $\Gamma = O^+(M)$. Then

$$\mathrm{QMod}_k(\Gamma) = \left(\mathrm{QMod}_k(\mathrm{SL}_2(\mathbb{Z})) \times \mathrm{QMod}_k(\mathrm{SL}_2(\mathbb{Z})) \right)^{\mathbb{Z}_2}.$$

Consider

$$F = R^2 \left(\frac{E_{10}}{\Delta} \right)$$

which has weight $k = 2$ but depth $d = 2$. Then $\mathrm{Lift}(F)$ is not almost-holomorphic.

On the other hand,

$$\mathrm{Lift}(G_2^*) = G_2^*(\tau_1)G_2^*(\tau_2)$$

is almost-holomorphic.

Another example:

$$H = R^d \left(\frac{E_{16}}{\Delta} \right)$$

which has weight $2d + 4$ but depth d , so $\mathrm{Lift}(H)$ is almost-holomorphic.

Suppose $M = U \oplus U(N) \oplus K(-1)$ where K is positive-definite,

Theorem

Let $F(\tau)$ be an almost-holomorphic modular form of weight $\kappa = k + 1 - \text{rank}(L)/2$ with $k \geq 0$. Consider the Fourier expansion of the constant term:

$$\text{ct}(F) = \sum_{\mu \in L^\vee / L} \sum_{n \in \mathbb{Q}} c^{(0)}(\mu; n) q^n \mathfrak{e}_\mu.$$

Assume the depth condition $k \geq 2d$ with $d > 0$. Then in open region of the tube domain

$$\text{ct}(\text{Lift}(F)) = \frac{1}{2} \zeta(1-k) c^{(0)}(0,0) + \sum_{\substack{\mu \in L^\vee \\ \mu > 0}} c^{(0)}(\mu, \mu^2/2) \sum_{\delta=1}^{\infty} \delta^{k-1} e^{2\pi i \langle \delta \mu, z \rangle}. \quad (2)$$

Enriques case

Recall the series which appeared as Gromov-Witten invariants of the local Enriques surface:

$$F_g = 2(-1)^{g-1} \sum_{\beta > 0} Q^\beta \sum_{\text{odd } k | \beta} k^{2g-3} \omega_g \left(\frac{\beta^2}{2k^2} \right)$$

where the coefficients $\omega_g(n)$ are defined by

$$\sum_{g \geq 0} \sum_{n \geq 0} \omega_g(n) z^{2g-2} q^n = \prod_{\substack{m \geq 1 \\ m \text{ odd}}} \frac{1}{(1 - e^z q^m)^2 (1 - e^{-z} q^m)^2 (1 - q^m)^4} \prod_{m \geq 1} \frac{1}{(1 - q^m)^8}.$$

Corollary

For any $g \geq 2$, F_g is a weak orthogonal quasimodular form of weight $2g - 2$ for the lattice $M = U \oplus U(2) \oplus E_8(-2)$ and group $O^+(M)$:

$$F_g \in \text{QMod}_{2g-2}^w(\Gamma).$$

Here "weak" is with respect to the classical Borcherds Enriques form $\Phi_4^2 \in \text{Mod}_8(\Gamma)$.

Let $f_g(\tau) = \sum_n \omega_g(n) q^n$. Then

$$f_g(\tau) = \left[\frac{\Theta(z, 2\tau)^2}{\Theta(z, \tau)^2} \right]_{z^{2g-2}} \cdot \frac{\eta(2\tau)^8}{\eta(\tau)^{16}}$$

where we used a normalized theta function.

$$\Theta(z) = \frac{\vartheta_1(z, \tau)}{\eta^3(\tau)} := (e^{z/2} - e^{-z/2}) \prod_{m \geq 1} \frac{(1 - e^z q^m)(1 - e^{-z} q^m)}{(1 - q^m)^2}.$$

It follows that f_g is a quasimodular form of depth $g - 1$ and weight $2g - 2$ for the group $\Gamma_0(2)$.

Consider $M = U \oplus U(2) \oplus E_8(-2)$. Then by a method of Scheithauer we can associate to a modular form $f \in \text{AHMod}_\kappa(\Gamma_0(2))$ a modular form F_f with respect to the Weil representation on $\mathbb{C}[M^\vee/M]$ by

$$F_f(\tau) := \sum_{A \in \Gamma_0(2) \backslash \text{SL}_2(\mathbb{Z})} (h|_\kappa A(\tau)) \rho_M(A^{-1}) \mathfrak{c}_0$$

Thus we can consider $\text{Lift}(F_{f_g^*})$. By our Theorem this is a weak almost-holomorphic modular form for $O^+(M)$. [The poles are along the Noether-Lefschetz divisors to square -2 classes, which is the zero locus of Φ_4^2 .] Then we still need to compute the Fourier expansion:

Proposition

Let $f \in \text{AHMod}_\kappa(\Gamma_0(2))$ be a modular form satisfying $k \geq 2d$ where $\kappa = k + 1 - \frac{n}{2}$. Let $f^{\text{Fr}}(\tau) := 2^{-\kappa/2} \tau^{-\kappa} f\left(-\frac{1}{2\tau}\right)$ be the Fricke involute. Consider the Fourier expansions of their constant terms $\text{ct}(f) = \sum_{n=0}^{\infty} d_n q^n$ and $\text{ct}(f^{\text{Fr}}) = \sum_{n=0}^{\infty} c_n q^n$. The lift $\text{Lift}(F_f)$ is a modular form with respect to $O^+(M)$. If moreover $c_{2n} + 2^{\frac{\kappa}{2}-1} d_n = 0$ for all n , then we have the expansion in the tube domain of $L(2) = U(2) \oplus E_8(-2)$:

$$\text{ct}(\text{Lift}(F_f)) = \frac{1}{2} 2^{(\kappa + \text{rk}(N))/2} \zeta(1-k) c_0 + \sum_{\ell \geq 1} \sum_{a, \alpha} q_2^\ell q_1^a \zeta^{\alpha} \sum_{\substack{\delta | (a, \ell, \alpha) \\ \gcd(\delta, 2) = 1}} \delta^{k-1} d_{\frac{a\ell}{\delta^2} - \frac{(\alpha, \alpha)}{2\delta^2}}. \quad (3)$$

where $q_1^a q_2^b \zeta^\alpha = e^{2\pi i a \tau_1} e^{2\pi i b \tau_2} e^{-2\pi i (\alpha, \mathfrak{z})_N}$ for (τ_1, τ_2) standard coordinates on U and $\mathfrak{z} \in E_8(-1)$.

III. Back to geometry

Let X be a smooth projective variety and $\beta \in H_2(X, \mathbb{Z})$.

Gromov-Witten invariants can be viewed as a collection of maps:

$$\langle \tau_{k_1, \dots, k_n}(-) \rangle_{g, \beta}^X : H^*(X^n) \rightarrow \mathbb{Q}.$$

for every g and $k_1, \dots, k_n \geq 0$ given by

$$\langle \tau_{k_1, \dots, k_n}(\gamma_1 \otimes \dots \otimes \gamma_n) \rangle_{g, \beta}^X = \int_{[\overline{M}_{g,n}(X, \beta)]^{\text{vir}}} \prod_{i=1}^n \text{ev}_i^*(\gamma_i) \psi_i^{k_i}$$

where $\overline{M}_{g,n}(X, \beta)$ is the moduli space of n -marked genus g stable maps $f : C \rightarrow X$.

For numerically K -trivial varieties the dimension of $[\overline{M}_{g,n}(X, \beta)]^{\text{vir}}$ does not depend on β and we can form generating series valued functions

$$F_g^X(\tau_{k_1, \dots, k_n}(-)) : H^*(X^n) \rightarrow \mathbb{Q}[[q^\beta]]$$

defined by

$$F_g^X(\tau_{k_1, \dots, k_n}(\Gamma)) := \sum_{\beta \geq 0} Q^\beta \langle \tau_{k_1, \dots, k_n}(\Gamma) \rangle_{g, \beta}^X$$

Conjecture (OW)

For Y an Enriques surface, every series $F_g^Y(\tau_{k_1, \dots, k_n}(\Gamma))$ is a weak quasimodular form for the lattice $M = U \oplus U(2) \oplus E_8(-2)$ and group $O^+(M)$ and some specific weight and rank.

Evidence:

- The computation of the F_g (=invariants of the local Enriques surface)
- True in genus 1
- Natural compatibilities are respected (holomorphic anomaly equation)

Open questions $F_2^Y(\tau_0(p)) \in \text{QMod}_4^w(\Gamma) = ?$

- Need better computational techniques.
- Description of spaces of quasimodular forms (in particular, vector-valued modular forms).

The Gromov-Witten invariants of a fixed K3 surface is zero by deformation invariance. This can be fixed by considering reduced invariants but this is not the right viewpoint in terms of modular properties. Instead we consider here invariants of families of K3 surfaces.

- Let L be a primitive sublattice of $U^3 \oplus E_8(-1)^2$ of signature $(1, r-1)$ with integral basis $L_1, \dots, L_r$.
- Let $\pi : \mathcal{X} \rightarrow B$ be a family of L -polarized K3 surfaces over a smooth base B . In particular, there is an embedding

$$L \subset \text{Pic}(\mathcal{X})_{\mathbb{Q}}$$

such that every fiber of π is a L -polarized K3 surface.

- Let $\overline{M}_{g,n}(\mathcal{X}/B, (d_1, \dots, d_r))$ be the moduli space of n -marked genus g stable maps to the fibers of π of degree $(d_1, \dots, d_r)$ against $L_1, \dots, L_r$.

The family Gromov-Witten invariants define a map:

$$\langle \tau_{k_1, \dots, k_n}(-) \rangle_{g, (d_1, \dots, d_r)}^{\mathcal{X}/B} : H^*(\mathcal{X}^n) \rightarrow H^*(B).$$

by essentially replacing the integral to the pushforward to the base B .

Formula in Codimension 1

Consider the case where the GW invariants of the family are divisors:

$$\langle \tau_{k_1, \dots, k_n}(\Gamma) \rangle_{g, (d_1, \dots, d_r)}^{\mathcal{X}/B} \in H^2(B).$$

Form the generating series

$$F_g(\tau_{k_1, \dots, k_n}(\Gamma)) = \sum_{d_1, \dots, d_r} q_1^{d_1} \cdots q_r^{d_r} \langle \tau_{k_1, \dots, k_n}(\Gamma) \rangle_{g, (d_1, \dots, d_r)}^{\mathcal{X}/B}$$

There is an explicit expression for these invariants:

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There is an explicit expression for these invariants:

Consider the generating series of Hegner/Noether-Lefschetz divisors of the family π :

$$\Phi^\pi(q) = \sum_n \sum_{\gamma \in L^\vee/L} \text{NL}_{n, \gamma} q^n e_\gamma$$

By a result of Borcherds, $\Phi^\pi(q)$ is a cycle-valued modular form for $\text{Mp}_2(\mathbb{Z})$ of weight $(22 - \text{rk}(L))/2$ for the Weil representation ρ_M on $M^\vee/M \cong (L^\vee/L)(-1)$ where $M = L^\perp$ is the orthogonal complement in the K3 lattice:

$$\Phi^\pi(q) \in H^2(B) \otimes \text{Mod}_{11 - \frac{\text{rk}(L)}{2}}(\text{Mp}_2(\mathbb{Z}), \rho_M).$$

Formula in Codimension 1

If Γ is a polynomial in diagonals and $c_2(S)$ pulled back from various factors, the reduced Gromov-Witten invariants of a K3 surface in primitive classes only depend on the square $\beta^2 = 2s$ of the curve class β . We write

$$F_g^{\text{red}}(\tau_{k_1, \dots, k_n}(\Gamma)) = \sum_{s \geq -1} \langle \tau_{k_1, \dots, k_n}(\Gamma) \rangle_{g,s}^{K3, \text{red}} q^s.$$

Theorem (Maulik-Pandharipande-Thomas, 2006)

$F_g^{\text{red}}(\tau_{k_1, \dots, k_n}(\Gamma))$ is a quasimodular form with simple pole at $q = 0$:

$$F_g^{\text{red}}(\tau_{k_1, \dots, k_n}(\Gamma)) \in \frac{1}{\Delta(q)} \text{QMod}_{2g + \deg_{\mathbb{C}}(\Gamma)}.$$

Set now

$$\Psi_{g, k_1, \dots, k_n, \Gamma}^{\pi} := \Phi^{\pi}(q) \cdot F_g^{\text{red}}(\tau_{k_1, \dots, k_n}(\Gamma)) \in H^2(B) \otimes \frac{1}{\Delta(q)} \text{QMod}(\text{Mp}_2(\mathbb{Z}), \rho_M).$$

which is of weight

$$\begin{aligned} \kappa &= 2g + \deg(\Gamma) - 12 + \left(11 - \frac{\text{rk}(L)}{2}\right) \\ &= 2g - 1 + \deg(\Gamma) - \text{rk}(L)/2 \end{aligned}$$

Theorem (O.-Pandharipande (2026, upcoming))

If the GW/PT correspondence holds for $(K3 \times \mathbb{P}^1, K3_{0,\infty})$ and $F_g(\tau_{k_1, \dots, k_n}(\Gamma))$ is a divisor, then

$$F_g(\tau_{k_1, \dots, k_n}(\Gamma)) = \sum_{\substack{\lambda \in L^\vee \\ \lambda > 0}} \sum_{a \geq 1} a^{2g-3+\deg(\Gamma)} \Psi_{g, k_1, \dots, k_n, \Gamma}^\pi [\lambda, (\lambda, \lambda)/2] Q^\lambda, \quad (4)$$

where $Q^\lambda = e^{2\pi i(\lambda, \mathfrak{z})L}$ where $\mathfrak{z} = \sum_i \mathfrak{z}_i L_i$ and $q_i = e^{2\pi i \mathfrak{z}_i}$.

Consider $U \oplus L$. Then the theta lift of $(\Psi_{g, k_1, \dots, k_n, \Gamma}^\pi)^*$ defines a function on the period domain of $U \oplus L$ of weight $k = 2g - 2 + \deg(\Gamma)$. If it were almost-holomorphic, then the Fourier expansion of its holomorphic part would be exactly (4). However, $\Psi_{g, k_1, \dots, k_n, \Gamma}^\pi$ has usually depth $g + \deg(\Gamma)/2$. Hence the degree bound

$$k = 2g - 2 + \deg(\Gamma) \geq \text{depth}(\Psi_{g, k_1, \dots, k_n, \Gamma}^\pi)$$

is usually not satisfied. Hence $\text{Lift}(\Psi_{g, k_1, \dots, k_n, \Gamma}^\pi)^*$ is not almost-holomorphic, and (4) is not quasimodular (with respect to the natural completion).

Question

- How to think about the functions $F_g(\tau_{k_1, \dots, k_n}(\Gamma))$?
- What is the modularity of the series $F_g(\tau_{k_1, \dots, k_n}(\Gamma))$ in higher codimension?

Thank you for your attention!